ECE 405: Introduction to large-scale AI systems Syllabus

This course is designed to give the technical background for a deep understanding of the problems associated with the development, deployment, and exploitation of Large Language Models, providing the students with a sufficient knowledge base to perform research directed at resolving these problems in academia or industry.
Aim: for students to gain the engineering capability to reproduce results from AI papers.

Instructor:

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Course website:

- The course webpage contains links to notes, recordings, and additional materials

Resources:

Main Stanford Course
Primer on Linear Algebra
John Hopkins course webpage
Cornell Tech course webpage
Secondary Stanford course webpage
Princeton course webpage

Pre-requisites:

linear algebra
statistics
calculus
basic ML
solid coding background (Python, PyTorch, bash, git)

Course Schedule

	Topics covered	Reading	Notes
Introduction			

Week 1: 13 Jan 2025 – 17 Jan 2025	Introduction. Course overview. Logistics. Language modeling. N-gram models. Tokenization: Byte-Pair Encoding		Problem set 1 released
Week 2: 20 Jan 2025 – 24 Jan 2025	Statistical modeling tools. Pytorch tensors and operations		
Week 3: 27 Jan 2025 – 31 Jan 2025	Pytorch Modules and optimizers. RNN. Self-attention		
Week 4: 3 Feb 2025 – 7 Feb 2025	Transformers and architecture variations.		
Week 5: 10 Feb 2025 – 14 Feb 2025	Self-supervised learning, quality filtering, and deduplication		Problem set 2 released
Language modeling			
Week 6: 17 Feb 2025 – 21 Feb 2025	Inference complexity computation. Inference latency. Inference throughput. Techniques for accelerated inference: Quantization, KV caching, FlashAttention, Speculative decoding		Problem set 1 deadline
Week 7: 24 Feb 2025 – 28 Feb 2025	Benchmarks: Automatic and human-based. Tasks: Classification, Question answering, summarization. Metrics: EM, F1, ROGUE, BLEU. Tools: HELM and HF Eval		
Week 8: 3 Mar 2025 – 7 Mar 2025	Supervised fine-tuning: large-scale and on budget		
Training			

Week 9: 10 Mar 2025 – 14 Mar 2025	Alignment to preferences: PPO, DPO, TPO		Problem set 3 released
Week 10: 24 Mar 2025 – 28 Mar 2025	Benchmarking, Profiling, CUDA, Triton		
Week 11: 31 Mar 2025 – 4 Apr 2025	Scaling laws, alpha parametrization, mu Parametrization		Problem set 2 deadline
Week 12: 7 Apr 2025 – 11 Apr 2025	Model, pipeline, context parallelism, synchronous training		Problem set 4 released
Week 13: 14 Apr 2025 – 18 Apr 2025	Rag, local inference with Ollama and vLLM		
Week 14: 21 Apr 2025 – 25 Apr 2025	Reasoning, tool use, agents		
Week 15: 28 Apr 2025 – 2 May 2025	Alternative architectures		Problem set 3 deadline
Week 16: 5 May 2025 – 9 May 2025	Merging modalities		Problem set 4 deadline

Logistics

The homework is due on Friday at midnight in the week listed in the course schedule.

There are a total of 5 late days available. You can use them for any of the assignments, but it is recommended that you keep them until later in the semester.

Assignment

The class will have four coding assignments on the main topics:

1. Basics of language modeling
    - Language model architecture (tokenization, transformer, sampling), pre-training pipeline, and hyperparameter ablations
2. Data
    - Data collection and pre-processing pipeline, quality filtering
3. Alignment
    - AI performance evaluation and methods for improving LLM performance: supervised fine-tuning on specific tasks, alignment to human preferences
4. Distributed computation
    - Benchmarking high-performance computational workflows and building distributed training pipelines.

Grade calculation

The grade will be based on the completed assignments (A1-A4).
Class grade = 0.25*A1 + 0.25*A2 + 0.25*A3 + 0.25*A4

Evaluation of the Course

Students' continued feedback is highly appreciated, both positive and negative. To that aim, a form will remain open throughout the
semester where you can provide feedback directly to me while remaining anonymous.
You can submit feedback at https://forms.gle/YFD1oAjpCD2wsTPg6. I will read every submission and take it into consideration.

There will also be an opportunity to provide formal feedback and evaluate the course at the end of the semester.

Course Policies

Inclusion

We are committed to creating a learning environment welcoming of all students that supports a diversity of thoughts, perspectives and experiences, and respects your identities and backgrounds (including race/ethnicity, nationality, gender identity, socioeconomic class, sexual orientation, language, religion, ability, etc.) To help accomplish this:

1. Please let us know if you have a name and/or set of pronouns that differ from those that appear in your official records.
2. If you feel like your performance in the class is being impacted by your experiences

outside of class (e.g., family matters, current events), please do not hesitate to come and talk with us. We want to be resources for you.

3. We (like many people) are still in the process of learning about diverse perspectives and identities. If something was said in class (by anyone) that made you feel uncomfortable, please talk to us about it.

As a participant in this class, recognize that you can be proactive about making other students feel included and respected.